SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Santosh(192521007) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Santosh(192521007)

3) Assume a real-time os used in embedded system controlling industrial machinery.

a) Type of Operating System

→ Real-Time Operating System (RTOS)

Reason:

* Fast Response
* Predictable execution.
* Meets deadlines
* Used in embedded and industrial systems

b) Diagram

User Process

kernel
Scheduler
Memory Manager
Device Driver

Hardware

c) Synchronization System Calls

* fork()
* wait()
* sem_wait()
* sem_post()
* pthread_mutex_lock()
* pthread_mutex_unlock()

These synchronize multiple process and prevent race conditions.

4) A distributed computing system consists of multiple interconnected computers sharing resources over a network.

a) Type of Operating System:

Distributed Operating System

Comparison

Distributed OS	Centralized OS
Multiple computers	Single computer
Shares resources	Local resources
High reliability	Less reliable
Faster resource sharing	Limited sharing

b) Distributed OS Architecture

Node 1 <---- Network ----> Node 2

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Kernel

Kernel

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Hardware

Hardware

c) IPC system call Sequence

pipe()
fork()
write()
read()
close()

or

socket()
bind()
listen()
accept()
send()
recv()
close()